

2.1 Consider the following five observations. You are to do all the parts of this exercise using only a calculator.

x	y	$x - \bar{x}$	$(x - \bar{x})^2$	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$
3	4	2	4	2	4
2	2	1	1	0	0
1	3	0	0	1	0
-1	1	-2	4	-1	2
0	0	-1	1	-2	2
$\sum x_i =$	$\sum y_i =$	$\sum (x_i - \bar{x}) =$	$\sum (x_i - \bar{x})^2 =$	$\sum (y_i - \bar{y}) =$	$\sum (x_i - \bar{x})(y_i - \bar{y}) =$
5	10	0	10	0	8

- a. Complete the entries in the table. Put the sums in the last row. What are the sample means $\bar{x}$ and $\bar{y}$?
b. Calculate b_1 and b_2 using (2.7) and (2.8) and state their interpretation.
c. Compute $\sum_{i=1}^5 x_i^2$, $\sum_{i=1}^5 x_i y_i$. Using these numerical values, show that $\sum (x_i - \bar{x})^2 = \sum x_i^2 - N\bar{x}^2$ and $\sum (x_i - \bar{x})(y_i - \bar{y}) = \sum x_i y_i - N\bar{x}\bar{y}$.

(a) $\bar{x} = \frac{1}{5} (3 + 2 + 1 - 1 + 0) = 1$
 $\bar{y} = \frac{1}{5} (4 + 2 + 3 + 1 + 0) = 2$

(b) $b_2 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = \frac{8}{10} = \frac{4}{5}$ $x \uparrow 1 \text{ unit} \Rightarrow y \uparrow \frac{4}{5} \text{ unit}$
 $b_1 = \bar{y} - b_2 \bar{x} = 2 - \frac{4}{5} \times 1 = \frac{6}{5}$ 當 $x=0$ 時, $y = \frac{6}{5}$

(c) $\sum_{i=1}^5 x_i^2 = 3^2 + 2^2 + 1^2 + (-1)^2 + 0^2 = 15$

$\sum_{i=1}^5 x_i y_i = 3 \times 4 + 2 \times 2 + 1 \times 3 + (-1) \times 1 + 0 \times 0 = 18$

$\sum_{i=1}^5 (x_i - \bar{x})^2 = 10 = 15 - 5 = \sum_{i=1}^5 x_i^2 - 5 \times \bar{x}^2$

$\sum_{i=1}^5 (x_i - \bar{x})(y_i - \bar{y}) = 8 = 18 - 10 = \sum_{i=1}^5 x_i y_i - 5 \times \bar{x} \bar{y}$

- d. Use the least squares estimates from part (b) to compute the fitted values of y , and complete the remainder of the table below. Put the sums in the last row.

Calculate the sample variance of y , $s_y^2 = \sum_{i=1}^N (y_i - \bar{y})^2 / (N - 1)$, the sample variance of x , $s_x^2 = \sum_{i=1}^N (x_i - \bar{x})^2 / (N - 1)$, the sample covariance between x and y , $s_{xy} = \sum_{i=1}^N (y_i - \bar{y})(x_i - \bar{x}) / (N - 1)$, the sample correlation between x and y , $r_{xy} = s_{xy} / (s_x s_y)$ and the coefficient of variation of x , $CV_x = 100(s_x / \bar{x})$. What is the median, 50th percentile, of x ?

x_i	y_i	$\hat{y}_i$	$\hat{e}_i$	$\hat{e}_i^2$	$x_i \hat{e}_i$
3	4	3.6	0.4	0.16	1.2
2	2	2.8	-0.8	0.64	-1.6
1	3	2	1	1	1
-1	1	0.4	0.6	0.36	-0.6
0	0	1.2	-1.2	1.44	0
$\sum x_i =$	$\sum y_i =$	$\sum \hat{y}_i =$	$\sum \hat{e}_i =$	$\sum \hat{e}_i^2 =$	$\sum x_i \hat{e}_i =$
5	10	10	0	3.6	0

$$\hat{y}_1 = \frac{4}{5}x_1 + \frac{6}{5}$$

$$s_y^2 = \sum_{i=1}^N (y_i - \bar{y})^2 / (N - 1) = 2.5$$

$$\hat{y}_1 = \frac{4}{5} \times 3 + \frac{6}{5} = \frac{18}{5} = 3.6$$

$$\hat{y}_4 = \frac{4}{5} \times (-1) + \frac{6}{5} = \frac{2}{5}$$

$$s_x^2 = 10 / 4 = 2.5$$

$$\hat{y}_2 = \frac{4}{5} \times 2 + \frac{6}{5} = \frac{14}{5} = 2.8$$

$$\hat{y}_5 = \frac{4}{5} \times 0 + \frac{6}{5} = \frac{6}{5}$$

$$s_{xy} = 8 / 4 = 2$$

$$\hat{y}_3 = \frac{4}{5} \times 1 + \frac{6}{5} = 2$$

$$r_{xy} = s_{xy} / (s_x s_y) = \frac{2}{\sqrt{2.5} \times \sqrt{2.5}} = 0.8$$

$$CV_x = 100 \times \left(\frac{s_x}{\bar{x}} \right) = 100 \times \frac{\sqrt{2.5}}{1} \approx 158.1139$$

50th percentile of $x = 1$

- e. On graph paper, plot the data points and sketch the fitted regression line $\hat{y}_i = b_1 + b_2 x_i$.

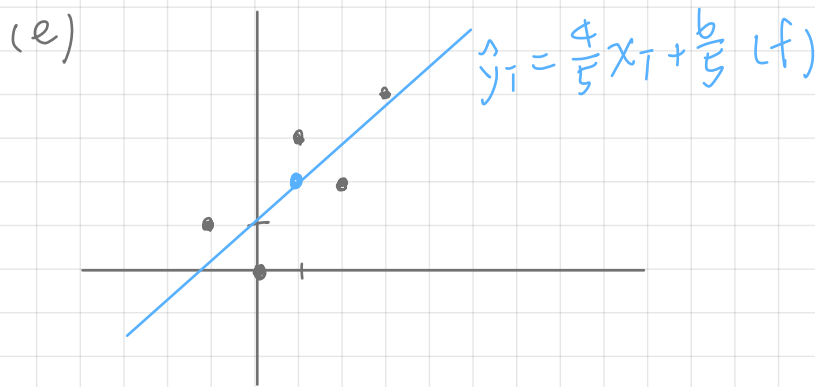
- f. On the sketch in part (e), locate the point of the means $(\bar{x}, \bar{y})$. Does your fitted line pass through that point? If not, go back to the drawing board, literally.

- g. Show that for these numerical values $\bar{y} = b_1 + b_2 \bar{x}$.

- h. Show that for these numerical values $\hat{y} = \bar{y}$, where $\hat{y} = \sum \hat{y}_i / N$.

- i. Compute $\hat{\sigma}^2$.

- j. Compute $\widehat{\text{var}}(b_2 | \mathbf{x})$ and $\text{se}(b_2)$.



$$(g) \bar{y} = 2 = \frac{b}{5} + \frac{4}{5} \times 1 = b_1 + b_2 \bar{x}$$

$$(h) \hat{\bar{y}} = \frac{1}{5} \left(\frac{18}{5} + \frac{14}{5} + 2 + \frac{2}{5} + \frac{6}{5} \right) = 2 = \bar{y}$$

$$(i) \hat{\sigma}^2 = \frac{\sum \hat{e}_i^2}{N-2} = \frac{\frac{18}{5}}{3} = 1.2$$

$$(j) \hat{\text{var}}(b_2|x) = \frac{b}{5} / 10 = \frac{3}{25}$$

$$\text{se}(b_2) = \sqrt{\hat{\text{var}}(b_2|x)} = \sqrt{\frac{3}{25}} \doteq 0.3464$$